Face Verification Using Siamese Networks: A Comparative Study

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1. Data Strategy

The LFW (Labeled Faces in the Wild) dataset contains 13,233 face images of 5,749 individuals. A significant challenge is the severe class imbalance: approximately 4,069 individuals have only a single image, making them unsuitable for positive pair or triplet construction.

Filtering Criteria: We retained only individuals with ≥ 2 images, resulting in 1,680 identities with 9,164 images. This threshold was chosen to

(i) enable positive pair/triplet generation,

(ii) maximize identity diversity for robust negative sampling,

(iii) avoid over-representation of a few heavily-photographed individuals (e.g., George W. Bush with 530 images).

Data Pipeline: We generated 5,000 training and 1,000 validation samples for each paradigm:

- **Pairs (Contrastive):** Balanced 50/50 positive (same person) and negative (different person) pairs. Positive pairs were created by randomly selecting 2 images of the same individual; negative pairs by sampling one image from each of two different individuals.
- **Triplets:** Each triplet consists of an anchor and positive image from the same person, with a negative image randomly sampled from a different identity.

All images were resized to 100×100 pixels and normalized to [0, 1].

2. Model Architecture Choices

Custom CNN Backbone

A 4-block convolutional architecture built from scratch with increasing filter depths (32 → 64 → 128 → 256). Each block uses two Conv2D layers with BatchNormalization and ReLU activations, followed by MaxPooling2D and Dropout (0.25). The embedding head consists of a 512-unit Dense layer, BatchNormalization, Dropout (0.5), and a 128-dimensional output with L2 normalization. Total trainable parameters: ~5.2M.

Transfer Learning Backbone (MobileNetV2)

MobileNetV2 pre-trained on ImageNet was used as a frozen feature extractor. Input images are internally rescaled from [0,1] to [-1,1] to match the expected distribution. A trainable embedding head (Dense 512 → BN → Dropout → Dense 128 → L2 norm) maps the 1,280-dimensional MobileNetV2 features to the 128-dimensional embedding space. Only ~790K parameters are trainable, with the 2.2M backbone parameters frozen.

Both architectures output L2-normalized embeddings, constraining vectors to a unit hypersphere for stable Euclidean distance computation (range: 0 to 2).

3. Loss Functions

Contrastive Loss

Manually implemented as: $L = y \cdot D^2 + (1-y) \cdot \max(0, m-D)^2$, where $y=1$ for same-person pairs, D is the Euclidean distance between embeddings, and $m=1.0$ is the margin. This directly optimizes absolute pairwise distances—pulling same-person embeddings closer and pushing different-person embeddings beyond the margin.

Triplet Loss

Manually implemented as: $L = \max(0, D(a,p)^2 - D(a,n)^2 + m)$, where $m=0.5$. This optimizes relative distance ordering, enforcing that anchor-positive distances remain smaller than anchor-negative distances by at least the margin.

4. Results & Critical Discussion

Contrastive vs. Triplet Loss

Contrastive loss demonstrated more stable convergence across both backbones. Its pair-based optimization provides direct supervision—each pair explicitly targets absolute distance minimization or maximization. **Triplet loss** showed higher variance during training, partly because random triplet sampling produces many "easy" triplets where the loss is already zero, yielding uninformative gradients. Hard negative mining (not implemented) would significantly improve triplet training by focusing on the most informative examples.

Custom CNN vs. Transfer Learning

The MobileNetV2 backbone benefited from hierarchical features learned on ImageNet's 1.4M images, providing strong initialization even for a different domain. It achieved faster initial convergence and competitive final performance with far fewer trainable parameters. The custom CNN required more epochs to develop meaningful feature representations but had the advantage of full end-to-end specialization for the face domain.

Distance Distribution Analysis

Well-trained models showed separation between positive and negative pair distance distributions. Greater separation corresponds to better verification performance. The optimal decision threshold, found by maximizing accuracy on the validation set, varied by model—typically ranging between 0.4 and 0.8 for L2-normalized embeddings.

Key Challenges

1. **Data scarcity:** Most LFW identities have very few images, limiting intra-class variation for positive pairs.
2. **Triplet sampling:** Random sampling is suboptimal; semi-hard mining would improve gradient quality.
3. **Margin sensitivity:** Both losses are sensitive to margin hyperparameters; values too small provide weak constraints while values too large make optimization difficult.

4. **Threshold calibration:** The verification threshold must be empirically tuned per model on a held-out set.

Future Directions

Angular margin losses (ArcFace, CosFace), online hard mining, data augmentation, and fine-tuning the last layers of MobileNetV2 would all improve performance further.